

Timbre Branch Architecture V1

As built specification for the explainable 128 to 35 concept bottleneck

Music Genre Classification Research Project
MTG Jamendo 7324 track subsample
Version 1 | September 2026

Decision The implemented timbre branch is accepted as Architecture V1. Its public contract is fixed at `h_audio [B,128]` to `z_timbre [B,35]`. The fusion module, not the timbre branch, owns the `Linear(35,64)` projection.

This document replaces the proposal's obsolete 35-to-32 alternative and its full-track extraction preference with the architecture, data policy, software contract, and evaluation procedure that are actually implemented.

Document status	Value
Architecture version	V1
Implementation status	Implemented and integrated
Branch input	Shared encoder pooled track representation [B,128]
Branch output	35 standardized named timbre concepts [B,35]
Fusion projection	Fusion owned <code>Linear(35,64)</code>
Dataset coverage	7324 unique tracks with 35 finite targets

1 Architecture verification

The source implementation and the original proposal agree on the central prediction head: 128 input values, two nonlinear hidden layers, and 35 unconstrained continuous outputs. The discrepancies were in the downstream interface and extraction policy, not in the neural head.

Design item	Original proposal	Implemented V1	Verdict
Input	Track embedding [B,128]	Track embedding [B,128]	Aligned
Hidden path	128 to 128 to 64	128 to 128 to 64	Aligned
Normalization and activation	LayerNorm, GELU	LayerNorm, GELU	Aligned
Dropout	0.20	0.20	Aligned
Concept output	35 continuous descriptors	35 standardized descriptors	Aligned
Branch projection	Optional 35 to 32	No branch owned token	Amended
Fusion projection	Unspecified or 32 wide	Fusion owned 35 to 64	Amended
Extraction scope	Full track preferred	First min(duration, 240 s)	Amended
Fusion bypass	Forbidden	Rejected by adapter	Aligned

1.1 Version 1 public contract

Input:	<code>h_audio</code>	<code>FloatTensor [B,128]</code>
Prediction:	<code>d_hat_standardized</code>	<code>FloatTensor [B,35]</code>
Published branch value	<code>z_timbre</code>	<code>FloatTensor [B,35]</code>
Supervision mask:	<code>valid_mask</code>	<code>BoolTensor [B,35]</code>
Fusion token:	<code>owned by fusion</code>	<code>FloatTensor [B,64]</code>

The branch does not return a learned fusion token and does not expose `h_audio` to fusion. The adapter rejects both violations. This preserves the semantic bottleneck.

2 Neural architecture

```
h_audio [B,128]
-> Linear(128,128)
-> LayerNorm(128)
-> GELU
-> Dropout(0.20)
-> Linear(128,64)
-> GELU
-> Linear(64,35)
-> z_timbre = d_hat_standardized [B,35]
```

Layer	Output shape	Role
Linear 128 to 128	[B,128]	Reorganizes the shared representation
LayerNorm and GELU	[B,128]	Stabilization and nonlinear mapping
Dropout 0.20	[B,128]	Regularization during training
Linear 128 to 64 and GELU	[B,64]	Internal timbre-oriented state
Linear 64 to 35	[B,35]	Named continuous concept bottleneck

The final layer has no sigmoid or softmax. Training targets are z-scored continuous values and may be positive or negative. The internal 64-dimensional activation is not an explanation and is never passed to fusion.

3 Timbre concept definition

The output order is a versioned part of the model contract. Extraction, standardization, checkpoint loading, fusion, evaluation, and explanation must use the same order.

Indices	Concept family	Features	Count
0 to 6	Spectral shape	Centroid mean and std; bandwidth mean and std; contrast mean; flatness mean; rolloff mean	7
7 to 8	Harmonic and noise	HNR mean in dB; inharmonicity mean	2
9 to 21	MFCC envelope level	MFCC 01 through 13 means	13
22 to 34	MFCC envelope variation	MFCC 01 through 13 standard deviations	13

3.1 Stable output names

```
spectral_centroid_mean, spectral_centroid_std
spectral_bandwidth_mean, spectral_bandwidth_std
spectral_contrast_mean, spectral_flatness_mean, spectral_rolloff_mean
hnr_mean_db, inharmonicity_mean
mfcc_01_mean ... mfcc_13_mean
mfcc_01_std ... mfcc_13_std
```

3.2 Meaning of the bottleneck

Every value reaching fusion through the timbre route corresponds to one declared acoustic descriptor. The model may learn correlated concepts, but it cannot send an unrestricted latent timbre vector around the 35 outputs. Explainability therefore rests on both the declared names and measured descriptor fidelity.

4 Dataset and extraction contract

Property	Version 1 decision
Track population	7324 selected MTG Jamendo tracks
Join key	TRACK_ID; row position is never used
Audio scope	First min(decoded duration, 240 seconds)
Target table	data/timbre_df.csv and corresponding columns in data/full_dataset.csv
Target width	35 finite continuous descriptors
Model input source	Shared encoder pooled 128-dimensional track representation
Split source	data/track_split_assignments.csv

The first-four-minute rule is the V1 experimental policy. It matches the audio span used to construct the selected log-Mel inputs and keeps branch supervision aligned with the shared encoder.

5 Preprocessing and leakage controls

1. Load the raw 35-value target vector by TRACK_ID.
2. Build the train, validation, and test membership from the official split manifest.
3. Fit one mean and scale per descriptor using training rows only.
4. Apply the saved training transformation unchanged to validation and test rows.
5. Retain a validity mask for every target cell and exclude invalid cells from loss computation.
6. Save feature order and scaler state inside the checkpoint.

$$d_standardized[:,j] = (d_raw[:,j] - train_mean[j]) / train_scale[j]$$

V1 uses ordinary per-feature z-score standardization. Log or robust transformations are not part of V1 and must receive a new preprocessing version if introduced later.

5.1 Loss

$$L_timbre = \text{sum}(\text{mask} * \text{SmoothL1}(d_hat, d_true)) / \text{max}(\text{sum}(\text{mask}), 1)$$

Smooth L1 limits the influence of occasional extraction outliers. A fully invalid batch returns a differentiable zero rather than fabricating target values.

6 Training and checkpoint contract

Setting	Implemented V1 default
Optimizer	AdamW
Learning rate	0.001
Weight decay	0.0001
Maximum epochs	50 in standalone timbre training
Batch size	128
Early stopping patience	8
Selection metric	Validation macro standardized descriptor MAE
Default seed	42

The standalone trainer restores the best validation checkpoint before one final test evaluation. The checkpoint stores format version, architecture name, model configuration, model state, feature order, training-only scaler, epoch, metrics, seed, and optimizer state.

7 Fusion integration

```
h_audio [B,128]
-> TimbreBranch
-> z_timbre [B,35]
-> TimbreAdapter
-> fusion-owned Linear(35,64)
-> timbre_token [B,64]
-> four-branch token stack [B,4,64]
```

The adapter accepts a raw [B,35] tensor or an inference dictionary containing z_timbre or d_hat_standardized. It validates dimensionality, finiteness, feature order, supervision mask, and fusion availability.

Rule	Enforcement
Timbre output is exactly [B,35]	Model configuration and adapter validation
No branch-owned fusion token	Adapter rejects fusion_token
No direct shared-embedding bypass	Adapter rejects h_audio in published output
Feature order cannot drift	Constants and checkpoint validation
Fusion creates the common width	TokenAssembler owns Linear(35,64)
Missing branch is masked	Fusion mask is applied after projection

7.1 Non-negotiable integrity rule

Fusion must never concatenate h_audio with z_timbre. Such a shortcut would allow genre prediction to bypass the named concepts and would turn the branch into an auxiliary regression head rather than a strict concept bottleneck.

8 Explanation outputs

Output	Shape	Purpose
z_timbre	[B,35]	Standardized tensor consumed by fusion
d_hat_standardized	[B,35]	Alias used for descriptor evaluation
d_hat_original_units	[B,35]	Inverse-scaled values for human-readable reports
feature_names	35 names	Stable semantic label for each dimension

9 Version 1 empirical checkpoint

The current joint test run provides evidence that the branch learns most declared concepts while retaining the fixed interface. These measurements describe the current checkpoint, not a guarantee for later training runs.

Metric	Result
Test standardized MAE	0.318
Test standardized RMSE	0.561
Test macro R squared	0.701
Observed target cells	38430 = 1098 tracks x 35 descriptors
Joint model test genre macro AP	0.736

Descriptor	Test R squared	Interpretation
spectral_contrast_mean	0.936	Very strong fidelity
mfcc_02_mean	0.931	Very strong fidelity
mfcc_01_mean	0.922	Very strong fidelity
spectral_rolloff_mean	0.881	Strong fidelity
spectral_flatness_mean	0.090	Weak; explanations require caution

The weak flatness result is documented rather than hidden. It motivates a future target-transform experiment, but V1 retains the original target definition and tensor contract.

10 Required research evaluations

- Report macro and per-descriptor MAE, RMSE, and R squared.
- Compare the complete model with a timbre-only model and a no-timbre ablation.
- Compare predicted concepts with oracle extracted concepts to measure the prediction gap.
- Run shuffled-concept and controlled concept-intervention tests.
- Report per-genre effects, not only aggregate genre scores.
- Repeat final comparisons over multiple seeds and report mean and dispersion.

11 Architecture version control

Architecture V1 is identified by its public tensor contract, descriptor order, preprocessing rules, and fusion ownership. Hyperparameter tuning that preserves these elements remains a V1 experiment.

Change	Version consequence
More epochs, early stopping, scheduler, or optimizer tuning	V1 training revision only
Different dropout or hidden width with [B,128] to [B,35] unchanged	V1 model revision; new checkpoint
Group-balanced loss	V1 training revision
Internal grouped output heads returning the same ordered [B,35]	V1.x internal architecture revision
Log-transforming one or more targets	New preprocessing version; document and checkpoint it
Adding or removing descriptors	New major architecture version
Changing input from pooled [B,128] to temporal [B,T,128]	New major architecture version
Passing h_audio directly to fusion	Not a valid strict bottleneck version

11.1 Version 1 acceptance tests

- A batch of h_audio with shape [B,128] returns finite values with shape [B,35].
- Incorrect input width and rank are rejected rather than silently reshaped.
- The published feature order contains exactly 35 unique names.
- The fusion adapter rejects a branch-owned fusion token and a direct h_audio bypass.
- The fusion projection maps [B,35] to [B,64] and respects the fusion mask.
- Gradients flow from genre and timbre losses through the 35-dimensional bottleneck.
- The scaler is fitted only on training rows and round-trips original acoustic units.
- Checkpoint reload reproduces predictions and validates the feature order.

12 Final V1 statement

The current implementation is accepted as Timbre Branch Architecture V1 after correcting the proposal's stale fusion and extraction descriptions. V1 is a strict learned concept bottleneck: the shared encoder supplies one 128-dimensional track representation, the timbre head predicts 35 standardized named descriptors, and fusion alone projects those concepts to a 64-dimensional branch token. No unrestricted audio representation bypasses the concepts.

Future performance improvements should preserve [B,128] input and [B,35] output unless the team explicitly declares and evaluates a new architecture version.